NEET UDAAN

Flight of Dreams

presents

Best Online Platform for

NEET - UG Preparation



STRUCTURE OF ATOM



DR. JYOTSANA | DR. ASHWANI | DR. RAJAT | DR. PRAVEEN | DR. PIYUSH | ER. NIKHIL | DR. SUDHIR

Structure Of Atom

Discovery

- Thomson's Model
- Rutherford's Model
- Bohr's Model
  - Results of Bohr's Model
  - Spectrum

De-Broglie Equation

Heisenberg Principle

Quantum Model

- Quantum Numbers
- Electronic Configuration
- Shape of Orbitals

1-2 Q/s

NEET UDAAN Flight of Dreams

Discovery

Thomson's Model
plum pudding model

positive sphere ← ooo ooo

electron

Rutherford's Model [α - particle scattering]
- Most of space is empty (NUCLEUS - DISCOVERY)
- Extremely small region of +ve charge at centre

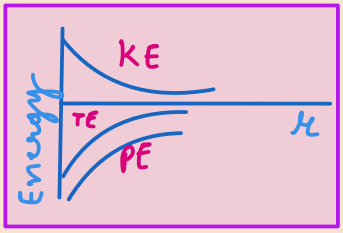
Why gold foil used ?
- Because of Larse atomic size and malleable nature.

Bohr's Model
- Electron revolve in stable orbits without loss of energy (Stationary Orbits)
- $L = \frac{nh}{2\pi}$ (L: Angular Moment of orbits is quantized)

RESULTS -
(Z = atomic No. of element)

- $V = 2.18 \times 10^6 \frac{Z}{n}$ m/s (velocity of electron in n^{th} orbit)
- $r = 0.529 \frac{n^2}{Z}$ Å (radius of orbit)

Neet 19 Odh



- K.E. of electron = $13.6 \frac{Z^2}{n^2}$ eV

  P.E. of electron = $-27.2 \frac{Z^2}{n^2}$ eV

  T.E. of electron = K.E. + P.E.

SPECTRUM -

No. of spectral lines $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

n_2 : higher energy level
n_1 : lower energy level

- **ABSORPTION**
electron transit from lower to higher energy level

- **EMISSION**
electron transit from higher to lower energy level

$$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$$

Wave number

Rydberg constant $1.0972 \times 10^7 m^{-1}$

Atomic No.

n_2 : higher energy level
n_1 : higher energy level

Lymann series If $n_2 = 1$
visible
Neet 19

Balmer series $n_2 = 2$

Paschen series $n_2 = 3$

Brackett series $n_2 = 4$

Pfund series $n_2 = 5$

Neet 2015 (P)

De Broglie Equation

Applicable for H and H like atom

6.62×10^{-34} J-S

$$\lambda = \frac{h}{p}$$

PMT-14

λ : wavelength
p : momentum

Heisenburg Uncertainity Principle

5.6×10^{-34} J-S

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Δx : change in position
Δp : change in momentum

Quantum Numbers (Q.N)

- PRINCIPAL Q.N. defines shell (n)
- AZIMUTHAL Q.N. defines subshell (ℓ)
- MAGNETIC Q.N defines orientation (m) (ORBITAL)
- SPIN Q.N defines spin of e⁻ (s)

dist. 2e⁻s

Neet 16

No. of subshells = n
No. of orbitals = n^2
No. of electron in shell = $2n^2$

No. of subshells = No. of ℓ values

$\ell = n-1$

$\ell = 0(s); 1(p); 2(d); 3(f)$

Orbital Angular Momentum

$\sqrt{\ell(\ell+1)}\ \frac{h}{2\pi}$

PMT-12

$(m = 2\ell+1)$ values

QUANTUM MODEL

Electronic Configuration

Rules -

- Pauli Exclusion Rule - NO 2 e⁻ can have same spin
- Afbau Rule - Filling acc. to increasing energy levels

$1s < 2s < 2p < 3s \ldots$

- Hund's Rule of Maximum Multiplicity

pairing of electrons occur only when they are filled singly

Cr : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$ $(Z=24)$

Cu : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$ $(Z=29)$

Shape of Orbitals

Radial Nodes — RN

- always spherical

$(n-l-1)$

ψ^2 = Probability Density

Angular Nodes — $= l$ — AN

- different shapes - mostly planar

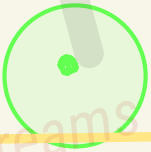
Total Nodes — $n-1$

Neet 2019

1s

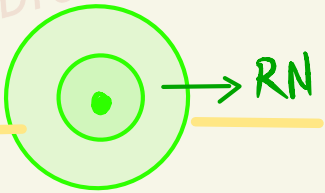
$RN = 1-0-1 = 0$
$AN = l = 0$

spherical



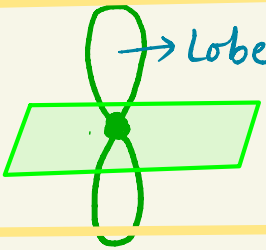
2s

$RN = 2-0-1 = 1$
$AN = l = 0$



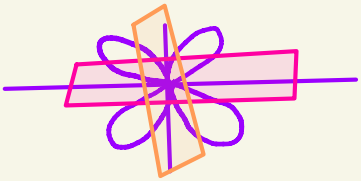
2p

$RN = 2-1-1 = 0$
$AN = l = 1$



3d

$RN = 3-2-1 = 0$
$AN = l = 2$



AN = 2 (in form of planes)

All the Best

WE BELIEVE IN YOU,
THAT YOU WILL NOT ONLY CRACK THE NEET EXAM....
BUT IN FUTURE YOU WILL ALSO CURE EACH
& VERY PATIENT WITH HONESTY..!!

WE ARE ALWAYS WITH YOU, JUST KEEP ON WORKING HARD...
ANY HELP YOU REQUIRE FROM OUR SIDE ;
NEVER HESITATE
TO ASK BECAUSE ALL OF YOU ARE LIKE
OUR OWN CHILDREN

from

NEET UDAAN
Flight of Dreams

Praveen Sir, Nikhil Sir, Jyotsana Mam, Sudhir Sir, Piyush Sir
[Teaching Faculty in NEET Udaan]